THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 7: **Proportional Reasoning-1** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Speed and time taken to cover a fixed distance are in: 1

(a) inverse proportion **(b)** direct proportion
(c) ratio 1 : 1 **(d)** no proportion

2. The ratio of 4 m 80 cm to 120 cm is: 1

(a) 5 : 1 **(b)** 4 : 1 **(c)** 1 : 4 **(d)** 4 : 2

3. If A : B = 4 : 5 and B : C = 10 : 9, then A : B : C is: 1

(a) 4 : 5 : 90 **(b)** 4 : 10 : 9 **(c)** 8 : 10 : 9 **(d)** 4 : 5 : 9

4. On a map, 1 cm represents 25 km. A distance of 2.4 cm on the map is actually: 1

(a) 60 km **(b)** 27.4 km **(c)** 6 km **(d)** 85 km

5. 10 workers can finish a job in 18 days. 15 workers will finish it in: 1

(a) 12 days **(b)** 13 days **(c)** 27 days **(d)** 14 days

6. A car covers 150 km in 2 hours at a steady speed. In 9 hours it covers: 1

(a) 84 km **(b)** 675 km **(c)** 159 km **(d)** 750 km

7. ₹ 1500 is divided among A, B and C in the ratio 2 : 5 : 8. The smallest share is: 1

(a) ₹ 200 **(b)** ₹ 500 **(c)** ₹ 800 **(d)** ₹ 300

8. In a proportion $a : b = c : d$, the product of the extremes equals: 1

(a) the sum of the means **(b)** the sum of the extremes
(c) the first term **(d)** the product of the means

9. The ratio of 2 m 25 cm to 50 cm is: 1

(a) 9 : 3 **(b)** 10 : 2 **(c)** 9 : 2 **(d)** 2 : 9

10. If A : B = 2 : 3 and B : C = 9 : 4, then A : B : C is: 1

(a) 2 : 3 : 4 **(b)** 6 : 9 : 4 **(c)** 2 : 3 : 36 **(d)** 2 : 9 : 4

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A map scale is 1 cm = 50 km. Find the actual distance for 2.6 cm and the map distance for 250 km. 2

12. Find the ratio of 1 m 80 cm to 45 cm in simplest form. 2

13. 7 workers complete a task in 24 days. How many days will 8 workers take? 2

14. A map scale is 1 cm = 40 km. Find the actual distance for 6.5 cm and the map distance for 200 km. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. If A : B = 2 : 3 and B : C = 9 : 4, find A : B : C. 3

16. 16 men can build a wall in 15 days. How many days will 10 men take to build it? 3

17. The ages of two friends are in the ratio 3 : 4 and the sum of their ages is 63 years. Find their ages after 3 years. 3

OR

If A : B = 5 : 6 and B : C = 12 : 7, find A : B : C. 3

18. 9 men can build a wall in 20 days. How many days will 6 men take to build it? 3

OR

The ages of two friends are in the ratio 7 : 8 and the sum of their ages is 30 years. Find their ages after 10 years. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Divide ₹ 3000 among A, B and C in the ratio 4 : 5 : 6. If C gives ₹ 400 to A, find the new ratio of their amounts. 5

OR

A bus covers 420 km in 6 hours at a uniform speed. Find 5
(i) its speed
(ii) the distance it covers in 4 hours
(iii) the time it takes to cover 210 km.

20. Divide ₹ 2100 among A, B and C in the ratio 1 : 2 : 4. If C gives ₹ 150 to A, find the new ratio of their amounts. 5

OR

A bus covers 270 km in 3 hours at a uniform speed. Find 5
(i) its speed
(ii) the distance it covers in 2 hours
(iii) the time it takes to cover 405 km.

END OF QUESTION PAPER — Total: 40 Marks

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